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Studierichting: Informatica
Oefeningenreeks 2E nr. 8.11

$$3^{x+4} \cdot 2^x = 0.0625 = \frac{1}{16}$$

$$\Rightarrow 3^{x+4} \cdot 2^x = \frac{1}{2^4} = 2^{-4}$$

$$\Rightarrow 3^x \cdot 3^4 \cdot 2^x = 2^{-4}$$

$$\Rightarrow 3^x \cdot 81 \cdot 2^x = 2^{-4}$$

$$\Rightarrow 81 \cdot (3 \cdot 2)^x = 2^{-4}$$

$$\Rightarrow 81 \cdot 6^x = 2^{-4}$$

$$\Rightarrow 6^x = \frac{2^{-4}}{81} = \frac{1}{16 \cdot 81} = \frac{1}{1296}$$

$$\Rightarrow 6^x = 6^{-4}$$

$$\Rightarrow x = -4$$

References